

ELK421E GENERATION OF ELECTRICAL ENERGY

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Number:					
Signature:					
Date: 12.01.2021					
Instructions +This Open Book examination paper consists of 3 questions. +You are not permitted to discuss or take assistance of others to solve the examination problems. +You would be severely penalized if your solutions appear to have been copied from others. +Your solutions should be methodical. You would be penalized if your solutions are illegible. +Please use your own statements while answering the questions					

Midterm (Open Book) Exam II

PROBLEM 1[30pts]: The 33 kV bus-bar of a station are in two sections A and B separated by a reactor. Section A is fed from four 10 MVA generators each having 20% reactance and section B is fed from the grid through 50 MVA transformer of 10% reactance. The circuit breakers have rupturing capacity of 500 MVA. Find the reactance of the reactor to prevent the circuit breakers being overloaded if a symmetrical short-circuit occurs on an outgoing feeder connected to it.

PROBLEM 2 [35pts]: A consumer has an annual consumption of 2×10^5 units. The tariff is 50 USD per kW of maximum demand plus 10 cent per kWh.

(i) Find the annual bill and the overall cost per kWh if the load factor is 35%.

(ii) What is the overall cost per kWh if the consumption were reduced by 25% with the same load factor ?

(iii) What is the overall cost per kWh if the load factor were 25% with the same consumption as in (i) ?

PROBLEM 3 [40pts]: Compare the annual cost of power supply to a factory having a maximum demand of 500 kW and a load factor of 40% by having the supply from:

(i) the factory's own diesel generating plant.

(ii) a public supply.

With regards to (i), the capital cost of factory's own generating plant is 800 000 USD, cost of fuel oil is 200 USD per ton, fuel consumption 295 gram per kWh. Capital charges, cost of repairs and maintenance, interest and depreciation 15% of the total capital cost. Salaries and wages of the operating staff are 15,000 USD per year.

With regards to (ii), the tariff is 150 USD per kW per annum of maximum demand plus 2.5 cent per kWh.

Which of the two alternatives is favourable for the operation of the factory [35pts]?

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Which of the two alternatives is favourable for the operation of the factory [35pts]?

$$\text{Max demand} = 500 \text{ kW}$$

$$LF = 0,4$$

$$\begin{array}{l} \text{interest} \\ \text{maintenance} \\ \text{depreciation} \end{array} \rightarrow 800\,000 \cdot 0,15 = 120\,000 \$$$

$$LF = \frac{\text{avg load}}{\text{max demand}}$$

$$\text{fuel} \rightarrow 200 \$ / \text{ton}$$

$$0,4 = \frac{\frac{\text{kWh}}{8760}}{500}$$

$$\text{kWh} = 1752000 \text{ kWh}$$

$$295 \text{ gr} / \text{kWh}$$

$$295 \cdot 1752000 = 516840000 \text{ gr}$$

$$516840000 \text{ gr} \rightarrow 516,84 \text{ ton}$$

$$200 \cdot 516,84 = 103368 \$$$

$$\text{Salaries} \rightarrow 15000 \$$$

$$\text{total cost} = 120000 + 103368 + 15000 = 238368 \$$$

$$\text{tariff} = 150 \$ / \text{kW} + 0,025 \$ / \text{kWh}$$

$$\text{total cost} = 500 \cdot 150 + 1752000 \cdot 0,025 = 418800 \$$$

PROBLEM 2 [35pts]: A consumer has an annual consumption of 2×10^5 units. The tariff is 50 USD per kW of maximum demand plus 10 cent per kWh.

(i) Find the annual bill and the overall cost per kWh if the load factor is 35%. $\text{overall } \$/\text{kWh}?$ $LF = 0,35$

(ii) What is the overall cost per kWh if the consumption were reduced by 25% with the same load factor? $\text{overall } \$/\text{kWh}?$ $2 \cdot 10^5 / 4$ units

(iii) What is the overall cost per kWh if the load factor were 25% with the same consumption as in (i)? $LF = 0,25$

a) $2 \cdot 10^5 \text{ units} \rightarrow 2 \cdot 10^5 \text{ kWh}$ $\text{Tariff} = 50 \$/\text{kW} + 0,1 \$/\text{kWh}$

$$LF = \frac{\text{avg load}}{\text{max demand}} \quad 0,35 = \frac{\frac{2 \cdot 10^5}{8760}}{\text{max demand}} \quad \text{max demand} = 65,23 \text{ kW}$$

$$\text{total cost} = 65,23 \cdot 50 + 2 \cdot 10^5 \cdot 0,1 = 23111,5 \$$$

$$\text{overall } \$/\text{kWh} = \frac{23111,5}{2 \cdot 10^5} = 0,11 \$/\text{kWh}$$

Salsora
kWh

b) $LF = 0,35 \quad \frac{3}{4} \cdot 2 \cdot 10^5 \text{ units} = 150000 \text{ kWh}$

$$0,35 = \frac{\frac{150000}{8760}}{\text{max demand}} \quad \text{max demand} = 48,92 \text{ kW}$$

$$\text{total cost} = 48,92 \cdot 50 + 150000 \cdot 0,1 = 17446 \$$$

$$\text{overall } \$/\text{kWh} = \frac{17446}{150000} = 0,11 \$/\text{kWh}$$

c) $LF = 0,25 \quad 2 \cdot 10^5 \text{ units} \rightarrow 2 \cdot 10^5 \text{ kWh}$

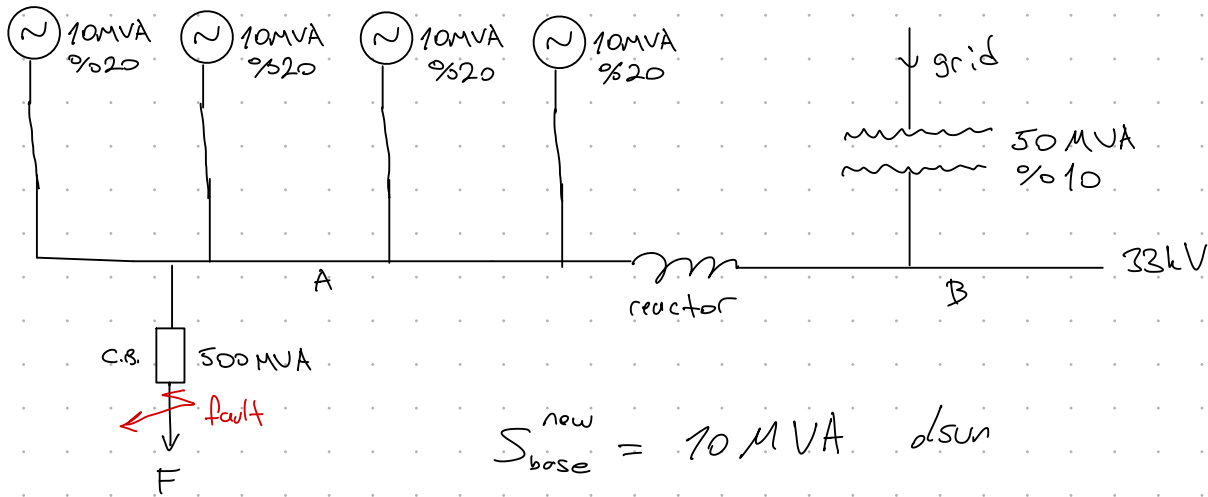
$$0,25 = \frac{\frac{2 \cdot 10^5}{8760}}{\text{max demand}} \quad \text{max demand} = 91,32 \text{ kW}$$

$$\text{overall cost} = 91,32 \cdot 50 + 2 \cdot 10^5 \cdot 0,1 = 24566 \$$$

$$\text{overall } \$/\text{kWh} = \frac{24566}{2 \cdot 10^5} = 0,122 \$/\text{kWh}$$

PROBLEM 1[30pts]: The 33 kV bus-bar of a station are in two sections A and B separated by a reactor. Section A is fed from four 10 MVA generators each having 20% reactance and section B is fed from the grid through 50 MVA transformer of 10% reactance. The circuit breakers have rupturing capacity of 500 MVA. Find the reactance of the reactor to prevent the circuit breakers being overloaded if a symmetrical short-circuit occurs on an outgoing feeder connected to it.

CB: X [A] ?

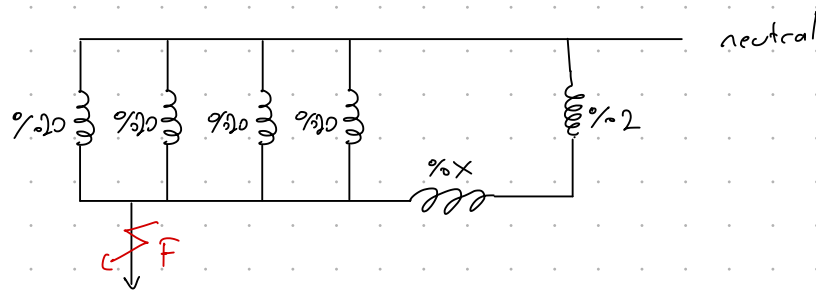


$$S_{base}^{new} = 10 \text{ MVA} \quad \text{dsun}$$

$$X_{base}^{new} = X_{base}^{old} \times \frac{S_{base}^{new}}{S_{base}^{old}}$$

$$\text{Generators} \rightarrow \%X = 20 \cdot \frac{10}{10} = \%20$$

$$\text{Transformer} \rightarrow \%X = 10 \cdot \frac{10}{50} = \%2$$



$$\text{eq. dger reactance} = \left(\%20 // \%20 // \%20 // \%20 \right) // (\%X + \%2)$$

$$= \frac{5 \cdot (X + 2)}{X + 7}$$

rupturing capacity

$$\text{Short-circuit kVA} = \text{Base kVA} \times \frac{100}{\%X}$$

Biz kiledik

$$500 = 10 \cdot \frac{100}{\frac{5 \cdot (X + 2)}{X + 7}}$$

$$0,5 = \frac{X + 2}{5X + 10}$$

$$2,5X + 5 = X + 2$$

$$1,5X = 2$$

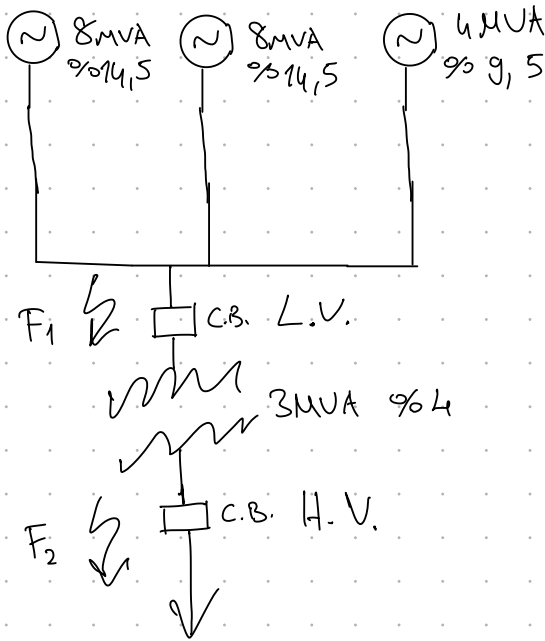
$$X = 1,33$$

$$\% \text{ reactance} = \frac{(\text{Base kVA}) \times \text{Reactance in } \Omega}{10 (\text{kV})^2}$$

$$1,33 = \frac{10 \cdot 10^3 \cdot R}{10 \cdot 33^2}$$

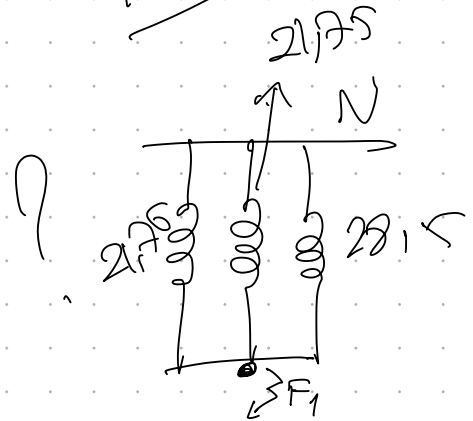
$$R = 1,44 \Omega$$

PROBLEM: The plant capacity of a 3-phase generating station consists of two 8 MVA generators of reactance 14.5% each and one 4 MVA generator of reactance 9.5%. These are connected to a common bus-bar from which loads are taken through a number of 3 MVA step-up transformers each having 4% reactance. Determine the MVA rating of the circuit breakers on (i) L.V. side and (ii) H.V. side. Reactances given are based on the MVA of each equipment



Section
new Base
12 MVA

S_{sc}



$$21.75 // 21.75 // 28.5$$

$$10.875 // 28.5$$

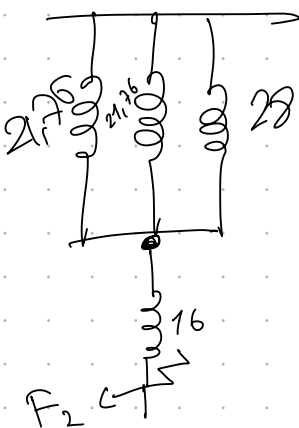
$$\%X = 7.871$$

$$\text{Short-circuit kVA} = \text{Base kVA} \times \frac{100}{\%X}$$

S_{sc} yani CB rating

(kVA) $\rightarrow \text{Base kVA} \cdot \frac{100}{\%X} = 12 \cdot 10^3 \cdot \frac{100}{7.871} = 152458 \text{ kVA}$

LV side



$$X_{base}^{new} = X_{base}^{old} \times \frac{S_{base}^{new}}{S_{base}^{old}}$$

$$\%X = 4 \cdot \frac{12}{3} = \%16$$

$$\text{total } X = 7.8 + 16 = \%23.8$$

$$\text{Short-circuit kVA} = \text{Base kVA} \times \frac{100}{\%X}$$

$$\text{Short circuit kVA} = 12 \cdot 10^3 \cdot \frac{100}{23.8} = 50420 \text{ kVA}$$

HG side

PROBLEM 4: The monthly readings of a consumer's meter are under:

Maximum demand = 60 kW

Energy consumed = 24000 kWh

Reactive energy = 15600 kVAR

If the tariff is \$20 per kW of maximum demand plus 3 cent per unit plus 0.1 cent per unit for each 1% power factor below 85%, **calculate the monthly bill of the consumer.**

1 ayddi \$?

$$\text{tariff} = 20\$/\text{kW} + 0.03\$/\text{kWh} + 0.001\$/\text{pf}$$

$$S = 60\text{k} + 15600\text{S} \quad |S| = 61994,83$$

$$\text{Pf} = \frac{60000}{61994,83} = 0,96 \rightarrow \text{so no bill for pf}$$

$$\begin{aligned} \text{total cost} &= 20 \cdot 60 + 0,03 \cdot 24000 = 1920\$ \text{ annual} \\ &\quad \underline{160\$ \text{ monthly}} \end{aligned}$$